

PH102: Tutorial 9 solution

1.

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_{\tau'} \frac{\vec{J}(\vec{r}')}{|\vec{r} - \vec{r}'|} d\tau' \quad (1)$$

In the given problem τ' is the volume of the shell and the origin of the coordinate system is at

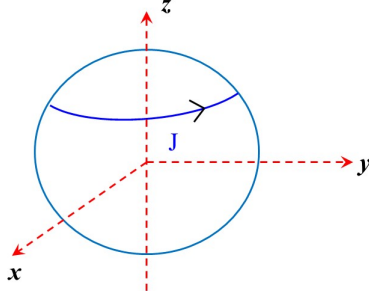


Fig.1

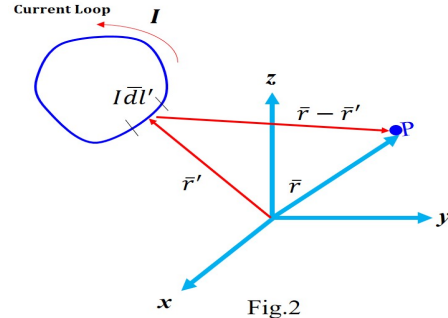


Fig.2

the center of the sphere, i.e., at $\vec{r} = 0$. The magnitude of $\vec{J}$ is uniform throughout the volume τ' of the spherical shell. Each current density element $\vec{J}d\tau'$ at a given point on the shell produces a differential vector potential $d\vec{A}$ at the center of the sphere, which is in the same direction as the current element. However, in view of the spherical symmetry, for each such element there exists another one located diametrically opposite and having oppositely directed current flow. This implies that the vector potential at the center of the sphere $\vec{A}(\vec{r} = 0) = 0$. On the other hand, $\vec{B}(\vec{r} = 0)$ does not vanish.

From the Biot Savart law we have,

$$\vec{B}(\vec{r} = 0) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}') \times \hat{R}}{(|\vec{r} - \vec{r}'|)^3} d\tau' \quad (2)$$

where, $\vec{J}(\vec{r}') = \hat{\phi} J_0$ is the current flowing between $r = a$ and $r = b$ and $\hat{R} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|}$ is the

unit vector pointing from a source point $\vec{r}'$ in the spherical shell to the origin and is the inward normal to the spherical surface. From symmetry, only the component $\vec{J} \times \hat{R}$ that is parallel to the polar axis of the sphere contributes to the $\vec{B}$ field at $\vec{r} = 0$. Noting that this component is proportional to $\sin \theta'$, where θ' is the spherical co-ordinate of the source point, we have

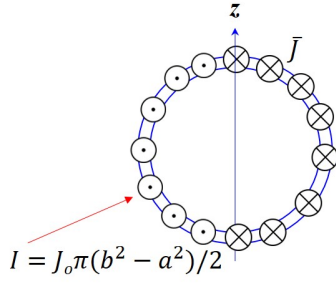
$$\vec{B}(\vec{r} = 0) = \frac{\mu_0}{4\pi} J_0 \int_{\tau'} \hat{z} \frac{\sin \theta'}{R^2} d\tau' \quad (3)$$

$$d\tau' = r'^2 \sin \theta' d\theta' d\phi' dr'$$

Thus,

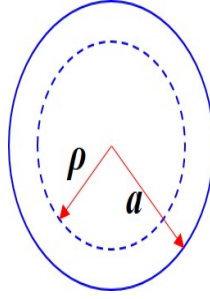
$$\begin{aligned} \vec{B}(\vec{r} = 0) &= \frac{\mu_0 J_0}{4\pi} \int_a^b dr' \int_0^\pi \sin^2 \theta' d\theta' \int_0^{2\pi} d\phi' \\ &= \frac{\mu_0 J_0 \pi^2}{4\pi} (b - a) \end{aligned}$$

If the shell carries a current I given by $I = \frac{\pi}{2} J_0 (b^2 - a^2)$, we can write



$$B_z(\vec{r} = 0) = \frac{\mu_0 I}{2(b + a)}$$

2. Applying Ampere's law we obtain



$$B_\phi 2\pi\rho = \mu_0 J_0 \pi \rho^2$$

$$\implies B_\phi = \frac{\mu_0 J_0}{2} \rho$$

However,

$$B_\phi = \vec{\nabla} \times \vec{A}|_\phi$$

$$= -\frac{\partial A_z}{d\rho}$$

$$\implies -\frac{dA_z}{d\rho} = \frac{\mu_0 J_0}{2} \rho$$

$$\implies A_z = -\mu_0 \frac{J_0}{4} \rho^2$$

Thus we finally have

$$\boxed{\vec{A} = -\frac{1}{4} \mu_0 J_0 \rho^2 \hat{z}}$$

3. Consider the annular strips of the disk of radius r and with dr . They will have a current

$$dI = \omega r \sigma dr,$$

where $\sigma = \frac{Q}{\pi R^2}$ is the surface charge density and ωr is the velocity of the strip. The current dI is the rate at which charge passes any point at radius r . The magnetic moment of the any strip is given by

$$dm = \pi r^2 dI$$

$$= \pi r^2 \omega r \sigma dr$$

$$= \pi r^2 \omega r \frac{Q}{\pi r^2} dr$$

$$= \frac{Q \omega r^3}{R^2} dr$$

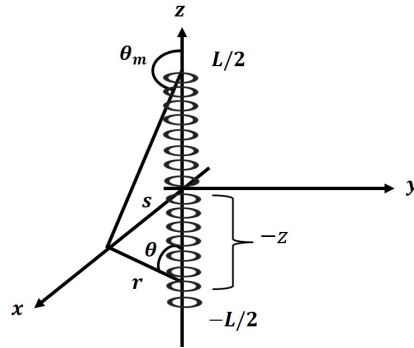
Thus the magnetic moment of the rotating disk is

$$\begin{aligned} m &= \frac{Q \omega}{R^2} \int_0^R r^3 dr \\ &= \frac{Q \omega R^2}{4} \end{aligned}$$

4. For a dipole at a origin and a field point in the xz plane ($\phi = 0$), we have

$$\begin{aligned} B &= \frac{\mu_0}{4\pi} \frac{m}{r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}) = \frac{\mu_0}{4\pi} \frac{m}{r^3} [2 \cos \theta (\sin \theta \hat{x} + \cos \theta \hat{z}) + \sin \theta (\cos \theta \hat{x} - \sin \theta \hat{z})] \\ &= \frac{\mu_0}{4\pi} \frac{m}{r^3} [3 \sin \theta \cos \theta \hat{x} + (2 \cos^2 \theta - \sin^2 \theta) \hat{z}] \end{aligned}$$

Here we have a stack of such dipoles, running from $z = -L/2$ to $z = +L/2$. From the figures we can see the $\hat{x}$ component of the magnetic field will get canceled out due to the symmetrical



arrangement of the dipoles above and below $z = 0$.

Therefore, we have

$$\vec{B} = \frac{\mu_0}{4\pi} 2\mathcal{M}\hat{z} \int_0^{L/2} \frac{(3\cos^2\theta - 1)}{r^3} dz$$

where $\mathcal{M}$ is the dipole moment per unit length. $m = I\pi R^2 = (\sigma v h)\pi R^2 = \sigma\omega R\pi R^2 h$, so we have $\mathcal{M} = \frac{m}{h} = \pi\sigma R^3$.

Now we have $\sin\theta = \frac{s}{r}$, so $\frac{1}{r^3} = \frac{\sin^3\theta}{s^3}$; $z = -s\cot\theta \implies dz = \frac{s}{\sin^2\theta}d\theta$. Therefore, the magnetic field will be

$$\begin{aligned} B &= \frac{\mu_0}{4\pi} (\pi\sigma\omega R^3) \hat{z} \int_{\pi/2}^{\theta_m} (3\cos^2\theta - 1) \frac{\sin^3\theta}{s^3} \frac{s}{\sin^2\theta} d\theta = \frac{\mu_0\sigma\omega R^3}{2s^2} \hat{z} \int_{\pi/2}^{\theta_m} (3\cos^2\theta - 1) \sin\theta d\theta \\ &= \frac{\mu_0\sigma\omega R^3}{2s^2} \hat{z} (-\cos^3\theta + \cos\theta) \Big|_{\pi/2}^{\theta_m} = \frac{\mu_0\sigma\omega R^3}{2s^2} \cos\theta_m (1 - \cos^2\theta_m) \hat{z} = \frac{\mu_0\sigma\omega R^3}{2s^2} \cos\theta_m \sin^2\theta_m \hat{z} \end{aligned}$$

But we have

$$\sin\theta_m = \frac{s}{\sqrt{s^2 + (L/2)^2}}$$

and

$$\cos\theta_m = \frac{-(L/2)}{\sqrt{s^2 + (L/2)^2}}$$

Therefore, finally we get

$$\boxed{B = -\frac{\mu_0\sigma\omega R^3 L}{4[s^2 + (L/2)^2]^{3/2}} \hat{z}}$$

5. Since the total energy is zero, we must have

$$-\sigma\pi r^2 + \sigma\pi(a^2 - r^2) = 0$$

$$\implies a^2 - 2r^2 = 0$$

$$\implies r = \frac{a}{\sqrt{2}}$$

The magnetic moment of a circular zone between radii r and $r + dr$ having a surface charge density σ is $\pi\omega r^3 dr$ (m=current X area= $\sigma v dr \times \pi r^2$; $v = \omega r$). The total magnetic moment of the rotating disc is therefore

$$\pi\sigma\omega \left[-\int_0^{a/\sqrt{2}} r^3 dr + \int_{a/\sqrt{2}}^a r^3 dr \right]$$

$$= \frac{\pi \sigma \omega a^4}{8}$$

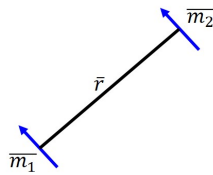
Thus,

$$\vec{m} = \frac{\pi \sigma a^4}{8} \vec{\omega}$$

Now, the uniform disc of mass M and radius a has a moment of inertia $\frac{1}{2}Ma^2$ about its figure axis. Thus the angular momentum of the rotating disc is $\vec{L} = \frac{1}{2}Ma^2\vec{\omega}$. Hence,

$$g = \frac{M}{L} = \frac{\pi \sigma a^4}{8} \frac{2}{Ma^2} = \frac{\pi \sigma a^2}{4M}$$

6. The magnetic field due to $\vec{m}_1$ at $r = L$ is



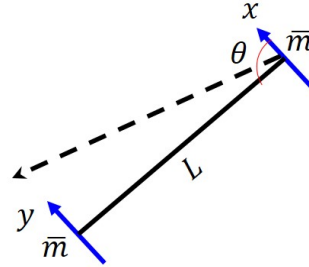
$$\vec{B}_1 = \frac{\mu_0}{4\pi} \left[\frac{3(\vec{m}_1 \cdot \hat{r})\hat{r}}{r^3} - \frac{\vec{m}_1}{r^3} \right]$$

Then the interaction energy of the two magnetic dipoles is given by

$$U = -\vec{m}_1 \cdot \vec{B}_1$$

$$= -\frac{\mu_0}{4\pi} \left[\frac{3(\vec{m}_1 \cdot \hat{r})(\vec{m}_2 \cdot \hat{r})}{r^3} - \frac{\vec{m}_1 \cdot \vec{m}_2}{r^3} \right]$$

Taking,



$$\vec{m}_1 = m_1 \hat{x}$$

$$\vec{m}_2 = m_2 \hat{x}$$

We obtain,

$$U = -\frac{\mu_0}{4\pi r^3} [3m_1 m_2 \cos^2 \theta - m_1^2]$$

Set

$$m_1 = m_2 = m$$

then

$$U = -\frac{\mu_0}{4\pi r^3} [3 \cos^2 \theta - 1]$$

Then,

$$\begin{aligned} \vec{F} &= -\vec{\nabla} U = \vec{\nabla} \left[\frac{\mu_0 m^2}{4\pi r^3} (3 \cos^2 \theta - 1) \right] \\ &= \frac{\mu_0 m^2}{4\pi} \left[\hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} \right] \left(\frac{3 \cos^2 \theta - 1}{r^3} \right) \\ \vec{F} &= \frac{\mu_0 m^2}{4\pi} \left[\hat{r} (3 \cos^2 \theta - 1) \left(-3 \frac{1}{r^4} \right) + \hat{\theta} \frac{1}{r^4} (-6 \cos \theta \sin \theta) \right] \\ &= \frac{\mu_0 m^2}{4\pi} \left[-3 \hat{r} \frac{(3 \cos^2 \theta - 1)}{r^4} - 6 \frac{1}{r^4} \cos \theta \sin \theta \hat{\theta} \right] \\ &= \frac{\mu_0}{4\pi} \frac{3m^2}{L^4} \left[(1 - 3 \cos^2 \theta) \hat{r} - \sin^2 \theta \hat{\theta} \right] \end{aligned}$$

$$\begin{aligned} \vec{\tau} &= \vec{m} \times \vec{B} \\ &= \frac{\mu_0}{4\pi} \vec{m} \times \left[\frac{3m \cos \theta}{r^3} \hat{r} - \frac{\vec{m}}{r^3} \right] \\ &= \frac{\mu_0}{4\pi} \frac{3m^2 \cos \theta}{r^3} (\hat{x} \times \hat{r}) \\ &= -\frac{\mu_0}{4\pi} \frac{3m^2 \cos \theta}{r^3} \hat{z} \\ &= -\frac{\mu_0}{4\pi} \frac{3m^2 \cos \theta}{L^3} \hat{z} \end{aligned}$$